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Drill Brain: A Patent-Pending Advisory Solution for Enhancing Drilling Performance Through Real Time Intelligence

A. Al Hai, O. Chammout, M. SharafEldin, A. Elbagoury, and K. Al Benali, ADNOC Onshore, Abu Dhabi, UAE

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Abstract

The increasing complexity and pace of drilling operations demand intelligent systems that offer proactive, data-driven solutions. At a Real-Time Intelligent center in Abu Dhabi, the Drill Brain, which is a patent-pending core advisory solution, is developed to support optimize drilling performance. It serves as a wholistic hub that query technical questions and provides recommendations to users based on historical and real time data. This platform will bridge AI to drilling operations, majorly enhancing the latter.

The Drill Brain is a grand platform hosting many models and solutions that are all integrated and connected. It acts as an advisory engine and a decision hub providing expert advice, proactive preventive measures, and best available practices based on both historical and current real time data processed at the hub. It offers knowledge repository, creating and maintaining a database which engineers can use to query technical questions to find solutions to drilling challenges. It also fully integrates with external data sources and applications to provide seamless workflow, for better enhancing of drilling operations and reduction of non-productive time.

The Drill Brain platform integrates AI and machine learning into the drilling process to facilitate smarter and faster decision-making and enhance operational efficiency. By using predictive algorithms to assess real time data, drilling teams can better anticipate potential issues, such as equipment failure, unforeseen geological conditions, or delays caused by non-optimized drilling parameters. This will not only enhance performance but also result in better utilization of resources, reducing downtime and optimizing the entire lifecycle of well development.

The Drill Brain captures and digests all drilling data, linking items together to provide actionable insights. It emulates the thought process of a drilling expert from natural language queries, in the form of a Chain of Thoughts (CoT), where each step is a standalone actionable drilling workflow. The CoT is automatically tailored to the using-entity's drilling data, tools, and expertise. The CoT's are also customizable to align with evolving workflows and capturing the knowledge of experienced users. It helps enhance the reasoning capabilities and consequently mitigate challenges related to local fields. This all results in saving rig durations, enhancing productivity, and ultimately maximizing ROI. This will all stream towards reducing the dollar value per drilled footage and thus enhance well delivery by 15% equivalent to over 350 million dollars.

Through real time intelligence, the novel concept of the Drill Brain is built on multiple systems and initiatives. The Drill Brain as a wholistic solution that envelops multi-discipline departments, AI tools, and advanced machine learning in a unified platform is something never done before. The conecpt is patent-pending and is utilized for first time in the onshore fields of Abu Dhabi.

Introduction

In recent years, the Real-Time Operations Center (RTOC) in Abu Dhabi has embarked on a remarkable and ambitious journey, transforming from its origins as a Real Time Monitoring Center (RTMC) into a state-of-the-art Real Time Operations Center. This evolution has fundamentally changed how drilling operations are envisioned, managed, and optimized. The transition was not merely a change of name or function, but a reimagining of the entire operational structure, marked by the integration of advanced technologies and innovative digital solutions (Jalbout et al, 2023).

The transformation journey paved the way for the seamless adoption of cutting-edge tools such as artificial intelligence, machine learning, and virtual reality. These technologies are being strategically embedded across diverse pillars of the organization, including People, Profitability, and Sustainability. By incorporating these digital advancements, the RTOC is strengthening its commitment to empowering its workforce, maximizing operational efficiency, and ensuring the sustainability of resources, all while maintaining strict alignment with the company's ambitious mindset (Benzine et al., 2024).

This bold approach is reshaping not only the technical aspects of well construction, but also the organizational culture and vision for the future. The focus has shifted from simply drilling wells to the broader and more sophisticated concept of "well manufacturing," where each phase is meticulously planned, simulated, and optimized in real time. Through this relentless pursuit of operational excellence and innovation, the RTOC is setting a new industry standard, one where the convergence of data-driven intelligence, digital infrastructure, and human expertise results in safer, smarter, and more sustainable energy production for years to come.

Scope

Well plans are defined by the operating company, often in consultation with the business partners. Experienced drillers and geologists study and review offset well data in order to create composite benchmarks. When the project kicks off, many challenges may arise during the execution of well operations, despite being properly planned. Monitoring and observing operations in real-time through the RTOC enable identifying additional shortfalls. These shortfalls are addressed by continuously adjusting the parameters and replanning based on real-time input. Such changes and adjustments may come at the cost of deviation from original plans and budgets. To control such possibilities and to further reduce costs, a strong process optimization is required for all stages, including real-time planning, monitoring, and execution.

Project Objectives

1. Enhance decision-making by providing real-time information on well conditions, equipment performance, and safety parameters
2. Reduce human intervention, enhance well construction performance, and maximize efficiency
3. Minimize risks by proactively addressing safety concerns during drilling operations
4. Enable quick analysis of drilling progress, downhole conditions, and potential issues
5. Reduce downtime and enhance rig productivity by intelligently managing schedules

Drill Brain - The Concept

The RTOC aims towards performance excellence through the full transition to the Real-Time Intelligent Center (RTIC). On trail of this visionary roadmap, the team is developing the following solutions to optimize planning, monitoring, and execution processes:

- **Drill Brain** - Core advisory solution that help query technical questions and provide recommendations to users based on historical and real-time data
- **Engineering Optimizer** - A comprehensive system looking after optimizing well designs during the planning phase based on historical data and offset wells analyses
- **Operations Optimizer** - A complete package aiming to dynamically and proactively enhance operations in real time all while tackling every possibility to achieve performance excellence

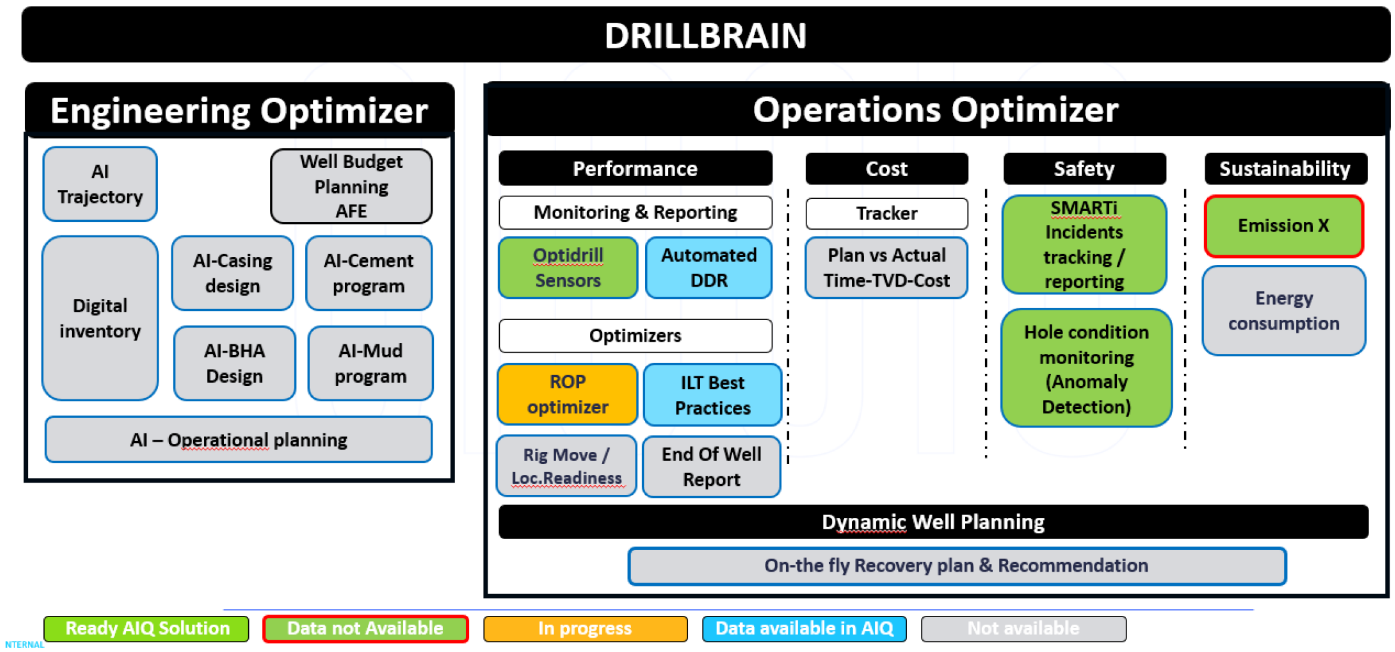

Figure 1—A layout of the Drill Brain concept

Drill Brain Models - The Concept Description

The Drill Brain is a grand platform hosting many models and solutions that are all integrated and connected. The Drill Brain acts as a:

- Advisory and Recommendation Engine: provide expert advice or troubleshooting support to personnel on past, planning and ongoing operations, real-time conditions, and best practices. Provide simulated drilling scenarios, helping understand complex situations like blowouts, stuck pipe incidents, or well control
- Knowledge Repository: create and maintain database which engineers can use to query technical questions to find solutions on drilling challenges
- Decision Hub: to support decision making by recommendations based on historical and real-time data from sensors to predict potential issues and optimize parameters to maximize efficiency and minimize wear on tools.
- Integration: full integration with external data sources and applications to provide seamless workflow, better optimizations and recommendations

The following is a description of each of the models and solutions available within Drill Brain.

AI Trajectory

The AI Trajectory module offers key functions such as recommending the optimal kick-off point based on well objectives, depth, and formation type, ensuring efficient wellpath initiation. It also proactively addresses Dogleg Severity (DLS) challenges by leveraging historical data and analyses from offset wells. By forecasting the impact of various drilling parameters on DLS, as well as considering Bottom Hole Assembly (BHA) specifications and limitations, the module helps optimize DLS in alignment with the selected BHA design, ultimately supporting safer and more effective drilling operations.

By analyzing historical and geological data, the system anticipates variations in formation and pressure, allowing for the prediction and mitigation of potential challenges such as kick risks, pressure surges, or unexpected hardening of formations. This approach enables the optimization of well trajectories based on offset data, enhancing safety and efficiency throughout the drilling process.

AI Casing Design

The AI Casing Design module provides several core functions, including optimization of casing selection by analyzing historical data and taking into account factors such as formation type, collapse risk, pressure, temperature, and fluid dynamics to maximize cost savings.

It also plays a significant role in risk mitigation by predicting potential issues with selected casing materials and anti-corrosion coatings, drawing from historical failure data that includes leaks, corrosion, and mechanical failures. Additionally, the module integrates with inventory systems to access information on casing availability, facilitate ordering, and streamline the optimization process.

AI Cement Program

The AI cement program encompasses several vital functions designed to enhance drilling efficiency and reliability. It addresses depth uncertainty by analyzing formation characteristics such as porosity and permeability, enabling the design of cement slurries and the evaluation of additives—like retarders, accelerators, and weight agents—to ensure effective performance under expected well conditions. By leveraging historical data, the system predicts crucial slurry properties including rheology, fluid loss, setting time, and flowability, recommending optimal combinations of additives for specific pressures and temperatures. The program also focuses on material optimization, promoting precise planning and execution to reduce waste and eliminate unnecessary cement use. AI-driven accurate displacement planning calculates optimal flow rates and volumes, ensuring thorough cement placement and effective mud removal. Additionally, the system simulates a range of cementing scenarios by varying well conditions, formation types, and cement properties to fine-tune the timing and placement of cement slurries, thus maximizing operational efficiency.

AI Mud Program

The AI Mud Program encompasses a range of functions aimed at optimizing mud properties and improving drilling operations. It enables custom mud formulation by fine-tuning variables such as density, viscosity, and fluid loss, all tailored to specific wellbore conditions and geological formations. By analyzing well logs, historical drilling data, and geological characteristics, the system enhances risk modeling and guides engineers in selecting optimal mud parameters to mitigate potential hazards. Automation features allow for the design of mud programs based on predefined parameters and previous successes, while the program also balances performance and cost by recommending the most cost-effective formulations for efficient well performance. Additionally, predictive maintenance capabilities monitor trends in the mud circulation system to identify and prevent possible failures, reducing the risk of unexpected operational disruptions.

AI BHA Design

The AI BHA Design module includes several key functions aimed at optimizing the bottom hole assembly for drilling operations. It analyzes geological, well trajectory, and operational data to recommend the most suitable tools, such as drill bits, stabilizers, reamers, motors, and measurement tools, tailoring BHA configurations to specific formations and objectives. The AI matches drill bit types and specifications with the BHA to maximize rate of penetration (ROP) and reduce wear, while also simulating and optimizing BHA configurations for balanced weight distribution, torque, and stability throughout drilling. Furthermore, the module simulates the mechanical behavior of shale formations under various stress conditions — including borehole pressure, wellbore stress, temperature variations, fracture toughness, and hydraulic stress —to anticipate and resolve shale instability challenges. Integrated inventory access also ensures real-time information on tool availability, ordering, and optimization, streamlining resource management within the drilling process.

Digital Inventory

Digital Inventory includes two main functions: planning and optimization. Planning relies on historical data, as well as factors like depth, geological conditions, and formations, to accurately determine the required inventory of drilling tools and materials, helping to prevent both overstocking and shortages. Optimization is achieved through analysis of vendor data, which enables the system to predict potential disruptions and recommend strategies such as maintaining backups of high-demand materials or components, or sourcing alternatives from other vendors or suppliers.

AI Operational Planning

Time prediction leverages historical data and operational parameters to forecast the duration of operations, enabling more effective planning and resource allocation. For risk mitigation, the system identifies potential hazards such as equipment failures, wellbore instability, or pressure issues by analyzing historical data and well logs, allowing for the optimization of key drilling parameters including weight on bit (WOB), rotary speed, and mud flow rates to enhance performance and reduce risks. Integration is achieved by analyzing information from diverse sources—such as geologists, engineers, and equipment suppliers—to promote collaboration and improve the quality of operations planning and decision-making.

Automated DDR

Real-time data streaming from various rig sensors ensures accurate and up-to-date reporting throughout operations. Automated report generation eliminates the need for manual intervention, streamlining the documentation process. Quality control is enhanced as AI algorithms detect data inconsistencies and identify gaps, thereby improving the overall quality and reliability of reporting. Additionally, video analysis provides valuable insights by processing video streams from the rig site, further supporting operational awareness and decision-making.

ILT Best Practices

Identifying best practices involves comparing offset wells to discover the most cost-effective drilling methods, enabling the standardization of procedures and a reduction in overall costs. Continuous improvement is achieved through regular post-job reviews and the systematic gathering of lessons learned. Additionally, implementing digital vision technologies allows computers to automate visual inspections, reducing reliance on manual checks and minimizing the potential for human error.

Rig Move / Location Readiness

Rig move sequencing and scheduling focus on determining the most efficient path and order for relocating one or multiple rigs and servicing several wells, all based on predefined criteria. The system automatically selects the best rig move mode to balance safety and cost-effectiveness, while real-time data is used to predict rig move release dates with greater accuracy. Additionally, equipment resource optimization is closely tied to location preparation and civil work, and a dynamic rig move schedule is maintained through integration with rig service provider.

End of Well Report

Automated data extraction streamlines the process of gathering relevant information from various sources, while template-based report generation ensures that standard templates are efficiently populated with this data. Additionally, retrospective analysis enables a thorough review of all operational data to identify, categorize, and analyze anomalies encountered during drilling or completion phases. This process not only assesses the impact of these anomalies on well performance but also offers recommendations for optimizing future wells.

Plan vs Actual Time-TVD-Cost

Effective real-time cost monitoring and control allows for the continuous tracking of expenses, enabling swift identification and correction of any cost overruns. Automated notifications highlight trends that could lead to overspending, ensuring proactive management. Comprehensive and transparent cost reports are generated, providing valuable comparisons between current drilling expenditures and historical data to reveal significant trends. Additionally, dynamic reporting ensures that updates in drilling operation costs are reflected promptly, while location-based tracking facilitates the comparison of multiple wells, supporting informed financial decision-making throughout the project.

Energy Consumption

Power management systems are implemented to prevent energy waste by automating operations based on expert knowledge, while real-time prediction capabilities estimate the energy required for upcoming activities, thereby helping to reduce gasoil consumption and lower CO_2 emissions. Additionally, smart control systems are used to optimize equipment operation, further minimizing overall energy consumption.

Dynamic Well Planning

Real-time optimization dynamically adjusts drilling plans whenever deviations from the original well path, anomalies, or unexpected changes are detected, thereby minimizing risks and keeping the project on target. At the same time, performance optimization and monitoring leverage AI to enhance drilling efficiency by continuously analyzing and refining operational parameters using real-time data. This approach not only optimizes equipment use and reduces downtime but also ensures the well remains aligned with project objectives through proactive, data-driven decision-making.

Results and Discussion

The Drill Brain Grand platform is forecasted to be fully operational in 2 years. The project road map is split into 2 main phases. Phase I will be building the foundations of the platform and achieving quick wins that helps verify the concept. After a trialing phase towards the end of year 2, the platform will go live and be scaled up to serve all drilling teams and sub-teams.

Project Milestones

The table below outlines the key deliverables and their main descriptions for the RTIC Project. These deliverables are subject to change after detailed scoping.

Table 1—A breakdown of the RTIC key deliverables

Key Deliverables	Description
Scoping & Planning	Completing & approving detailed scoping, project documentation, resource planning and project timeline
Platform Development & Data Integration	Platform development & Integration with existing ADNOC Onshore data sources
Drill Brain development & Implementation	Development and training Drill Brain on existing AIQ solutions
Operations Optimization (AI)	Development of the remaining modules for Operations Optimization with the use of ML/AI Modelling and connecting / training Drill Brain on new modules
DDR Automation	DDR / data input automation using ML to automate daily drilling reports
Engineering Optimization (Drill Brain Connectivity)	Connecting & training Drill Brain on existing applications and available data for Engineering Optimization solution.
Engineering Optimization (ML/AI Adjustment)	Adjusting & optimizing existing solution with the use of ML/AI to improve Drill Brain

Business and Financial Gains

The Drill Brain platform through the new and revamped RTIC will achieve major economic benefits. Based on data analytics exceeding P50 of the historical data, it is calculated that the Drill Brain will help achieve the following gains:

- 2% NPT avoidance
- 5% ROP optimization
- 2% Design validation
- Risk assessment
- 2% Cost optimization
- Safety
- 3% Well planning
- Material optimization
- 2% ILT savings

All of the above will stream towards reducing the dollar value per drilled footage ($/ft) and consequently enhancing well delivery and business plans by an astonishing **15%** equivalent to over $100 million . The true goal is to further exceed the limits and achieve the P75 targets, which will result in greater well delivery and generate accelerated business revenues.

Conclusion

The RTIC is significantly enhancing wells performance, driving powerful results through the "Drill Brain" by fully scaling up the integration of AI and digitalization. Built on four sturdy pillars of digitalized HSE, Well Planning, Well Engineering, and Well Operations, the RTIC steers maximum optimization through divulging the invisible areas of improvement. Digitalized HSE elevates safety, using real-time data to predict and prevent risks, creating a safer and more sustainable work environment. With Digitalized Well Planning, RTIC enables accurate and efficient business planning for the whole Drilling Function. Digitalized Well Engineering redefines the process of making well programs, transforming it to full automation, for cost effective and efficient well delivery. Finally, Digitalized Well Operations enhances drilling and completions performance through automated proactive alerting of anomalies minimizing downtime, and maximizing performance via AI prediction tools. RTIC not only makes drilling operations become smarter but also set a new standard for sustainable and efficient energy production.

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Nomenclature

RTMC : Real Time Monitoring Center
RTOC : Real Time Operations Center
RTIC : Real Time Intelligent Center
AI : Artificial Intelligence
ML : Machine learning
VR : Virtual reality
WOB : Wait on bit
SPP : Standpipe pressure

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